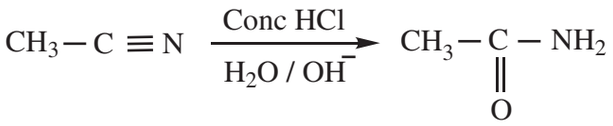
3) Partial hydrolysis of alkyl cyanides (Nitriles)

Partial hydrolysis of alkyl cyanides with cold con HCl gives amides



Methyl cyanide — Acetamide

Chemical Properties

1. Amphoteric character

Amides behave both as weak acid as well as weak base and thus show amphoteric character. This can be proved by the following reactions.

Acetamide (as base) reacts with hydrochloric acid to form salt



Acetamide — Acetamide hydrochloride

Acetamide (as acid) reacts with sodium to form sodium salt and hydrogen gas is liberated.



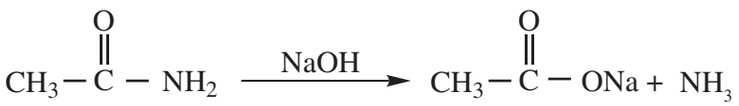
Acetamide — Sodium acetamide

2) Hydrolysis

Amides can be hydrolysed in acid or in alkaline solution on prolonged heating



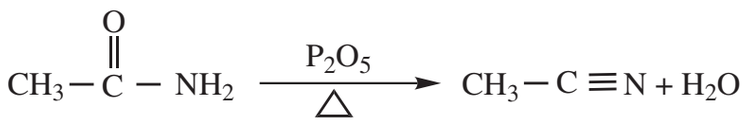
Acetamide — Acetic acid



Acetamide — Sodium acetate

3) Dehydration

Amides on heating with strong dehydrating agents like P_2O_5 get dehydrated to form cyanides.



Acetamide — Methyl cyanide (aceto nitrile)

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